SOLUTION

SECTION–I

1. Ans. (C)

Sol. E : none of the drawn balls are green
E_1 : 2R balls are drawn
E_2 : 2W balls are drawn
E_3 : 1R + 1W balls are drawn

$$P(E_1/E)$$

$$=\frac{P(E_1).P(E/E_1)}{P(E_1)P(E/E_1)+P(E_2).P(E/E_2)+P(E_3).P(E/E_3)}$$

$$=\frac{\frac{{}^3C_2}{{}^7C_2}\times 1}{\frac{{}^3C_2}{{}^7C_2}\times 1+\frac{{}^2C_2}{{}^7C_2}\times 1+\frac{{}^3C_1\times{}^2C_1}{{}^7C_2}}=\frac{3}{10}$$

2. Ans. (C)

Sol. $P(A)-P(A\cap B)=\frac{1}{3}$ &

$$P(A\cup B)=P(A)+P(B)-P(A\cap B)=\frac{11}{15}$$

$$\Rightarrow \quad P(B)=\frac{6}{15}=\frac{2}{5}$$

$$P(A)-P(A)\,P(B)=\frac{1}{3}\Rightarrow P(A)=\frac{5}{9}$$

$$\Rightarrow \quad P(A\cap B)=P(A)\,P(B)=\frac{2}{5}\times\frac{5}{9}=\frac{2}{9}$$

3. Ans. (B)

Sol. $\because \quad \{P\}=P \Rightarrow P\in[0,1)$
$\Rightarrow \quad P\neq 1$
$P(E)\neq 1 \Rightarrow$ E is not a sure event.

4. Ans. (A)

Sol. T : took a taxi
R : Took a rickshaw
W : Walked by foot

$$P(T)=\frac{1}{2},P(R)=\frac{1}{3},P(W)=\frac{1}{6}$$

C : Catching the train

$$P(C/R)=\frac{1}{2}\ P(C/T)\ ;\ P(C/W)=\frac{1}{4}P(C/R)$$

let $P(C/R)=x$

$$\therefore\ P(C/T)=2x\ ;\ P(C/W)=\frac{x}{4}$$

$$P(W/C)=\frac{\frac{1}{6}\times\frac{x}{4}}{\frac{1}{2}\times 2x+\frac{1}{3}\times x+\frac{1}{6}\times\frac{x}{4}}$$

$$=\frac{\frac{1}{24}}{1+\frac{1}{3}+\frac{1}{24}}=\frac{1}{24+8+1}=\frac{1}{33}$$

5. Ans. (D)

Sol. All throws are independent

$$\therefore\ P=\frac{1}{6}$$

6. Ans. (C)

Sol. $P(A\cap\bar{C})=P(A\cap\bar{C}\cap B)+P(A\cap\bar{C}\cap\bar{B})$

$$=\frac{1}{3}+\frac{1}{3}=\frac{2}{3}$$

$$P(A)=P(A\cap C)+P(A\cap\bar{C})$$

$$\Rightarrow P(A\cap C)=\frac{3}{4}-\frac{2}{3}=\frac{1}{12}$$

7. Ans. (A)

Sol. $P(i)=ki$

$$\because \quad \sum_{i=1}^{6}P(i)=1 \Rightarrow k(1+2+3+4+5+6)=1$$

$$\Rightarrow \quad k=\frac{1}{21}$$

$$\therefore \quad P(\text{even})\times P(\text{odd})$$

$$=\left(\frac{2}{21}+\frac{4}{21}+\frac{6}{21}\right)\times\left(\frac{1}{21}+\frac{3}{21}+\frac{5}{21}\right)=\frac{12}{49}$$

8. Ans. (B)

Sol. P(WBWB) + P(BWBW)

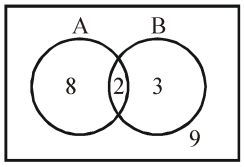
$$=\frac{5}{8}\times\frac{3}{8}\times\frac{5}{8}\times\frac{3}{8}+\frac{3}{8}\times\frac{5}{8}\times\frac{3}{8}\times\frac{5}{8}$$

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$$= 2\left(\frac{15}{64}\right)^2 = \frac{15^2}{2^{11}}$$

9. Ans. (B,C,D)

Sol. $P(\bar{A}) = \frac{6}{11}; P(A) = \frac{5}{11}$



$$P(B) = \frac{5}{22},\ P(A \cap B) = \frac{1}{11}$$

$$P(A \cup B) = \frac{13}{22},\ P(\bar{A} \cap \bar{B}) = \frac{9}{22}$$

$$P(\bar{A} \cup \bar{B}) = \frac{10}{11},\ P(\bar{B} \cap A) = \frac{8}{22} = \frac{4}{11}$$

10. Ans.(A,B,C,D)

Sol. (A) $P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = 1 - P(A \cap B)$

$$1 - P(A)\,P(B) = \frac{5}{6}$$

(B) $= P(B) - P(B \cap C) = P(B) - P(B)P(C)$

$$= \frac{5}{18}$$

(C) $= P(\overline{A \cup B}) = 1 - P(A \cup B)$

$$= 1 - P(A) - P(B) + P(A \cap B)$$

$$= 1 - P(A) - P(B) + P(A)P(B) = \frac{1}{3}$$

(D) $= P(B) - P(A \cap B) + P(C) - P(B \cap C) + P(A) - P(A \cap C)$

$$= P(B) - P(A)\,P(B) + P(C) - P(B)\,P(C) + P(A) - P(C)\,P(A) = \frac{25}{36}$$

11. Ans.(A,B,C,D)

Sol. (A) $P(A \cup B \cup C) = 1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$

$$1 - \frac{2}{3}.\frac{1}{2}.\frac{3}{4} = \frac{18}{24} = \frac{3}{4}$$

(B) $\Sigma P(AB) - 3P(ABC) = \frac{1}{6} + \frac{1}{8} + \frac{1}{12} - \frac{3}{24}$

$$= \frac{4+3+2-3}{24} = \frac{1}{4}$$

(C) $\Sigma P(AB) - 2P(ABC) = \frac{1}{6} + \frac{1}{8} + \frac{1}{12} - \frac{2}{24} = \frac{7}{24}$

(D) $1 - P(A \cap B \cap C) = 1 - \frac{1}{3} \times \frac{1}{2} \times \frac{1}{4} = \frac{23}{24}$

Paragraph for Question 13 to 14

13. Ans. (C)

Sol. $P(B) = \frac{5}{6}p_1 + \frac{5}{6}(1-p_1)(1-p_2).\frac{5}{6}p_1 + \left(\frac{5}{6}\right)^3 (1-p_1)^2(1-p_2)^2 p_1 + \ldots\ldots$

$$P(B) = \frac{\frac{5}{6}p_1}{1 - \frac{5}{6}(1-p_1)(1-p_2)},\ P(C) = \frac{\frac{5}{6}(1-p_1)p_2}{1 - \frac{5}{6}(1-p_1)(1-p_2)}$$

$$\frac{P(B)}{P(C)} = \frac{p_1}{p_2(1-p_1)}$$

14. Ans.(C)

Sol. If game is equally likely
$\Rightarrow$ Probability becomes

$\frac{1}{3}$ for each player $\Rightarrow \frac{P(B)}{P(C)} = 1$

$\Rightarrow p_1 = p_2(1-p_1)$ ...(i)

$$\& \ P(A) = \frac{\frac{1}{6}}{1 - \frac{5}{6}(1-p_1)(1-p_2)} = \frac{1}{3}$$

$\Rightarrow (1-p_1)(1-p_2) = \frac{3}{5}$ ...(ii)

solve (i) & (ii) $\Rightarrow p_2 = \frac{1}{4}, p_1 = \frac{1}{5}$

$\Rightarrow p_1 : p_2 = 4:5$

Paragraph for Question 14 & 15

14. Ans. (A)

Sol. P(student will pass) $= \frac{1}{7}.\frac{1}{2} + \frac{1}{3}.\frac{1}{3} + \frac{11}{21}.\frac{1}{6} = \frac{17}{63}$

15. Ans. (C)

Sol. P(y/ student is fail) $= \dfrac{\frac{1}{3}.\frac{2}{3}}{\frac{1}{7}.\frac{1}{2} + \frac{1}{3}.\frac{2}{3} + \frac{11}{21}.\frac{5}{6}} = \frac{7}{23}$

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SECTION–II

1. Ans. 2.00

Sol. Prime numbers

{2, 3, 5, 7, 11, 13, 17, 19, 23, 29}

P(atleast one) = 1 – P(no prime)

$$=1-\frac{^{20}C_2}{^{30}C_2}=1-\frac{20.19}{30.29}=\frac{49}{87}$$

$$\therefore \quad \frac{b-a}{19}=\frac{87-49}{19}=2$$

2. Ans. 106.00

Sol. Two balls transferred from I to II can be 1W and 1B or both W or both B.

Then the required probability

$$=\left(\frac{^4C_1\,^3C_1}{^7C_2}\right)\cdot\frac{6}{9}+\left(\frac{^4C_2}{^7C_2}\right)\frac{7}{9}+\left(\frac{^3C_2}{^7C_2}\right)\cdot\frac{5}{9}$$

$$=\frac{43}{63}=\frac{p}{q} \quad \therefore\ p+q=106$$

3. Ans. 4.00

Sol. Possible points that Humpty Not scores are

$$=\begin{cases}9 & \text{i.e. } 333\\ 8 & \text{i.e. } 233\\ 7 & \text{i.e } 223\end{cases}$$

Required probability = 1–P(he score 7,8 or 9)

$$=1-\{^3C_2(0.3)^2.(0.5)+^3C_2.(0.5)^2.(0.3)+^3C_3.(0.5)^3\}$$

$$=1-\left\{\frac{3.9.5+3.25.3+1.125}{1000}\right\}=\frac{103}{200}$$

4. Ans. 0.49

Sol. E_r : hunters shoots the animal at r distance

P(E) = P(escapes) = 1 – P $(\bar{E}_{2a}\bar{E}_{3a}.....\bar{E}_{na})$

$$=1-\left(1-\frac{a^2}{4a^2}\right)\left(1-\frac{a^2}{9a^2}\right)....\left(1-\frac{a^2}{n^2a^2}\right)$$

$$=1-\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)....\left(1-\frac{1}{n^2}\right)$$

$$=1-\left(\frac{1.3}{2.2}\right)\left(\frac{2}{3}.\frac{4}{3}\right)\left(\frac{3}{4}.\frac{5}{4}\right)....\left(\frac{n-1}{n}\frac{n+1}{n}\right)$$

$$=1-\frac{n+1}{2n}=\frac{n-1}{2n}=\frac{49}{100}=0.49$$

SECTION–IV

1. Ans. (A)→(P); (B)→(Q,T); (C)→(P); (D)→(S,T)

Sol. (A) Let number of black balls = x

$$\Rightarrow \quad \frac{^2C_1.^2C_1+^2C_1.^xC_1+^2C_1.^xC_1}{^{x+4}C_2}=\frac{5}{7}$$

$$\Rightarrow \quad 7(4+4x)=\frac{5(x+4)(x+3)}{2}$$

$$\Rightarrow \quad 56 + 56x = 5x^2 + 35x + 60$$

$$\Rightarrow \quad 5x^2 - 21x + 4 = 0$$

$$\Rightarrow \quad (5x - 1)(x - 4) = 0 \quad \Rightarrow \quad x = 4$$

(B) $\because$ A,B are independent

$\Rightarrow$ $A, \bar{B}$ are also independent

$$\Rightarrow \quad P(A/\bar{B})=P(A)=\frac{1}{3}$$

(C) 1 – P(1 on none of the throw) > 0.5

$$\Rightarrow \quad 1-\left(\frac{5}{6}\right)^n>0.5 \Rightarrow \quad \left(\frac{6}{5}\right)^n>2$$

$$\Rightarrow \quad n_{min} = 4$$

(D) P(3H & 5T) = $^8C_3\times\left(\frac{1}{2}\right)^3\times\left(\frac{1}{2}\right)^5$

$$=56\times\frac{1}{8\times32}$$

$$\Rightarrow \quad \frac{7}{32}$$

$\Rightarrow$ odds in favour = 7 to 25

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